



HF Controls Ltd
100 Albert Drive, Glasgow, G41 2SJ

www.hf-group.co.uk
Call 0800 783 9480

Glasgow Edinburgh Manchester Belfast Aberdeen

PROJECT: RSGT
PROJECT LOCATION: LIVERPOOL
PROJECT REFERENCE NUMBER: 291926

CLIENT: PEEL PORTS
CLIENT ADDRESS: MARITIME CENTRE PORT OF LIVERPOOL
CLIENT ORDER NUMBER: ????

CONTROL SYSTEM DETAILS & SPECIFICATION

SYSTEM TYPE:	SAFE AREA
ENCLOSURE MATERIAL:	MILD STEEL
RATED VOLTAGE:	230V AC
RATED FREQUENCY:	50 Hz
RATED AMPERE:	6A
CONTROL VOLTAGE:	24V DC
DEGREE OF PROTECTION:	IP 55
PAINT FINISH:	POWDER COATED RAL7035
YEAR OF MANUFACTURE:	2022



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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	M.WRIGHT
00	12/01/24	AS FITTED			DRAWN	M.WRIGHT
01					CHECKED	D.BALDOCK
02					DATE	17/03/2023
03			INSTALL	RSGT LIVERPOOL	DRG. No.: 284595_2101	
04						
05						
06						
07			TITLE	CP02 IS PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:
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A	B	C	D	E	F																																																
<table><tr><td>CURRENT CIRCUIT</td><td>CONDUCTOR</td><td>WIRING COLOUR</td><td>MINIMUM CROSS-SECTION</td></tr><tr><td>3-PHASE POWER</td><td>L1</td><td>BROWN</td><td>2.5mm2</td></tr><tr><td>3-PHASE POWER</td><td>L2</td><td>BLACK</td><td>2.5mm2</td></tr><tr><td>3-PHASE POWER</td><td>L3</td><td>GREY</td><td>2.5mm2</td></tr><tr><td>3-PHASE NEUTRAL</td><td>N</td><td>LIGHT BLUE</td><td>2.5mm2</td></tr><tr><td>1-PHASE POWER</td><td>L1</td><td>BROWN</td><td>1.5mm2</td></tr><tr><td>1-PHASE POWER</td><td>N</td><td>LIGHT BLUE</td><td>1.5mm2</td></tr><tr><td>24VDC</td><td>+</td><td>DARK BLUE</td><td>0.75mm2</td></tr><tr><td>24VDC</td><td>-</td><td>DARK BLUE</td><td>0.75mm2</td></tr><tr><td>MEASURING</td><td>+/-</td><td>GREY</td><td>0.5mm2</td></tr><tr><td>EXTERNAL VOLTAGE</td><td></td><td>ORANGE</td><td>1.5mm2</td></tr><tr><td>PROTECTIVE EARTH CONDUCTOR</td><td>PE</td><td>GREEN/YELLOW</td><td>1.5mm2</td></tr></table>						CURRENT CIRCUIT	CONDUCTOR	WIRING COLOUR	MINIMUM CROSS-SECTION	3-PHASE POWER	L1	BROWN	2.5mm2	3-PHASE POWER	L2	BLACK	2.5mm2	3-PHASE POWER	L3	GREY	2.5mm2	3-PHASE NEUTRAL	N	LIGHT BLUE	2.5mm2	1-PHASE POWER	L1	BROWN	1.5mm2	1-PHASE POWER	N	LIGHT BLUE	1.5mm2	24VDC	+	DARK BLUE	0.75mm2	24VDC	-	DARK BLUE	0.75mm2	MEASURING	+/-	GREY	0.5mm2	EXTERNAL VOLTAGE		ORANGE	1.5mm2	PROTECTIVE EARTH CONDUCTOR	PE	GREEN/YELLOW	1.5mm2
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MEASURING LINES:

- ☒ SINGLE CONDUCTOR
☐ TWISTED PAIRS
☒ ANALOGUE SIGNAL IN SHIELDED PAIRS

TYPE OF CONNECTION:

- ☒ WIRE END BOOTLACE CRIMP
☐ WIRE END PIN CRIMP
☐ FORK TYPE CRIMP
☐ ISOLATED
☐ NO CRIMP

WIRE MARKING:

- ☐ NONE
☐ WIRE DESIGNATION, SEMI OPEN
☐ WIRE TARGET DESTINATION
☒ CURRENT PATH DESIGNATION

SPECIFICS:

- ☐ HALOGEN FREE
☒ IP20 FOR INFEED TERMINALS
☒ BUS-BAR PE

CONSTRUCTION NOTES:

- | | |
|---------------|--------|
| 1. TERMINALS: | |
| POWER | WDU4 |
| INSTRUMENT | WDU2.5 |



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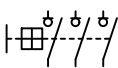


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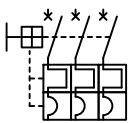
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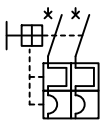
CODE OF ELECTRICAL SYMBOLS



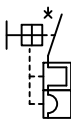
Q = MAIN ISOLATING SWITCH



QFM = ISOLATING CIRCUIT BREAKER
(3 POLE)



QFM = ISOLATING CIRCUIT BREAKER
(2 POLE)



QFM = ISOLATING CIRCUIT BREAKER
(1 POLE)



CONTACT (POWER)
NORMALLY OPENED



CONTACT-THERMOSTAT
NORMALLY CLOSED



FU = OVERLOAD PROTECTIVE DEVICE
(3 POLE)



SF = SELECTOR SWITCH
2 POSITION / STAY PUT



SF = PUSH BUTTON / SPRING RETURN
NORMALLY OPEN



SF = PUSH BUTTON / SPRING RETURN
NORMALLY CLOSED



Kx = CONTROL COIL



PLC INPUT



PLC OUTPUT



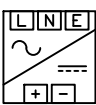
INDICATION LED



HEATER



RESISTOR



PSU = AC TO DC POWER CONVERSION



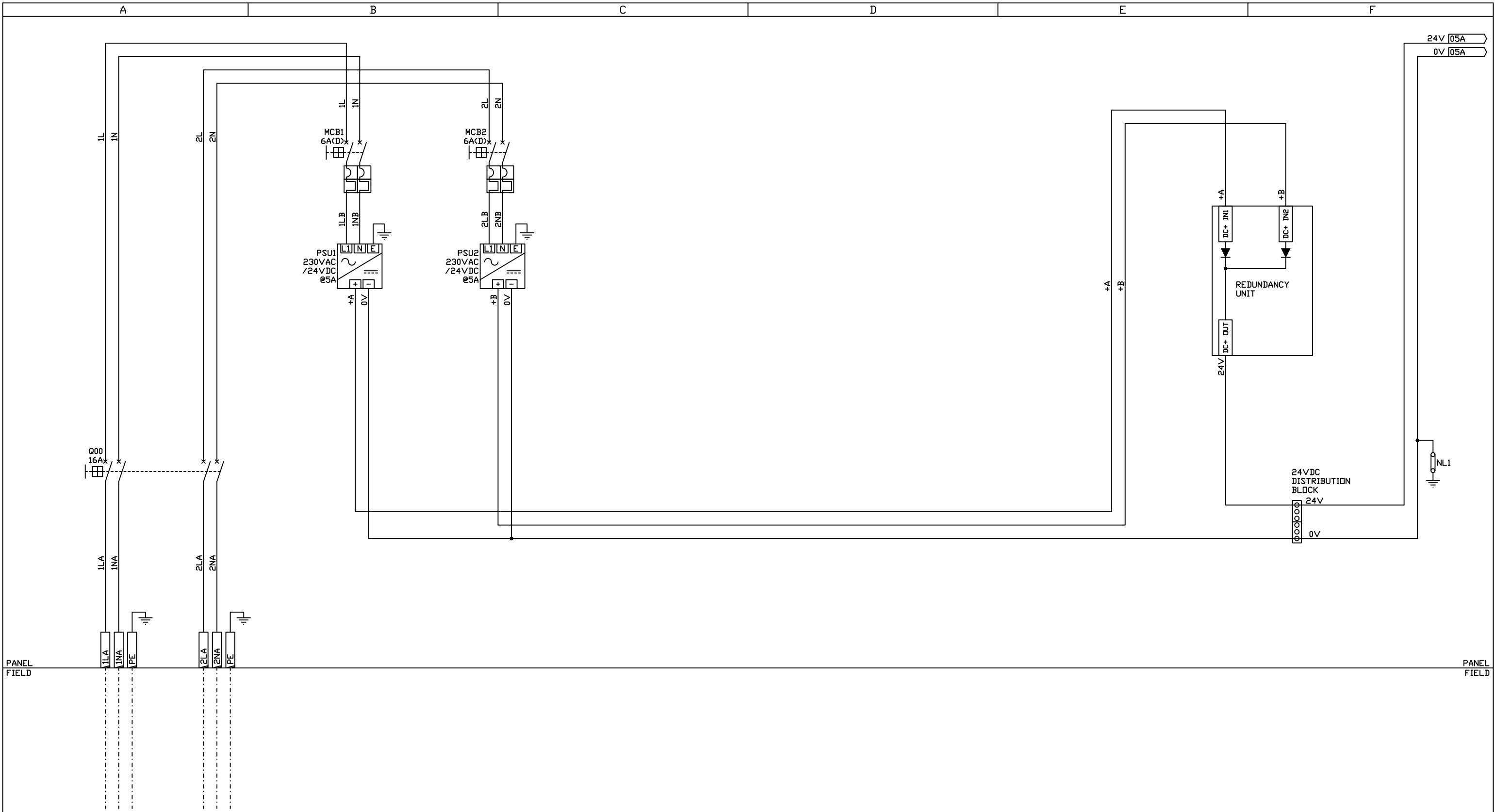
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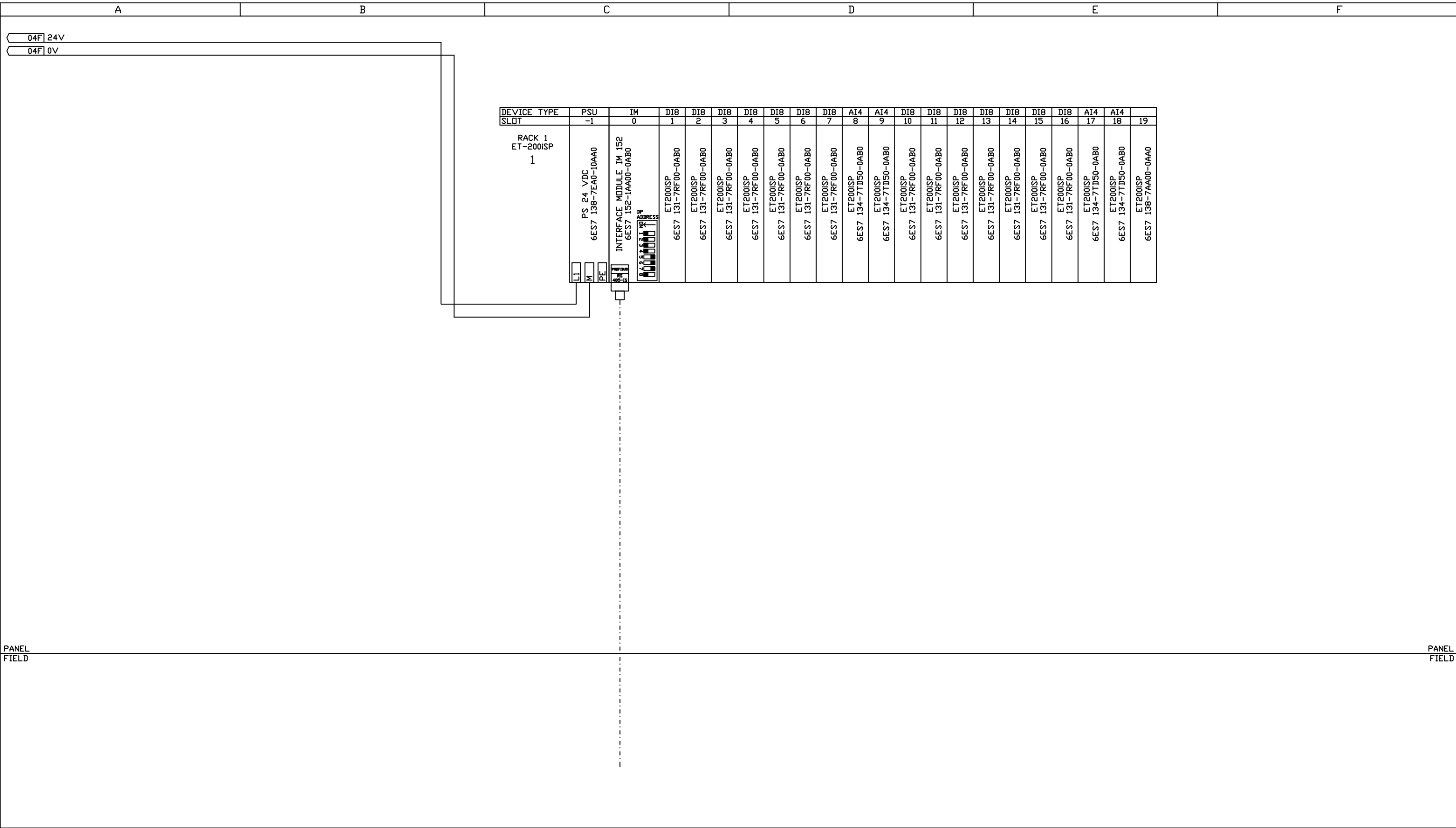
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	MAINS INCOMING #1 SUPPLY 1PH/N/E 230VAC 6A	MAINS INCOMING #2 SUPPLY 1PH/N/E 230VAC 6A		24VDC POWER SUPPLY 1		24VDC POWER SUPPLY 2			24VDC REDUNDANCY UNIT							
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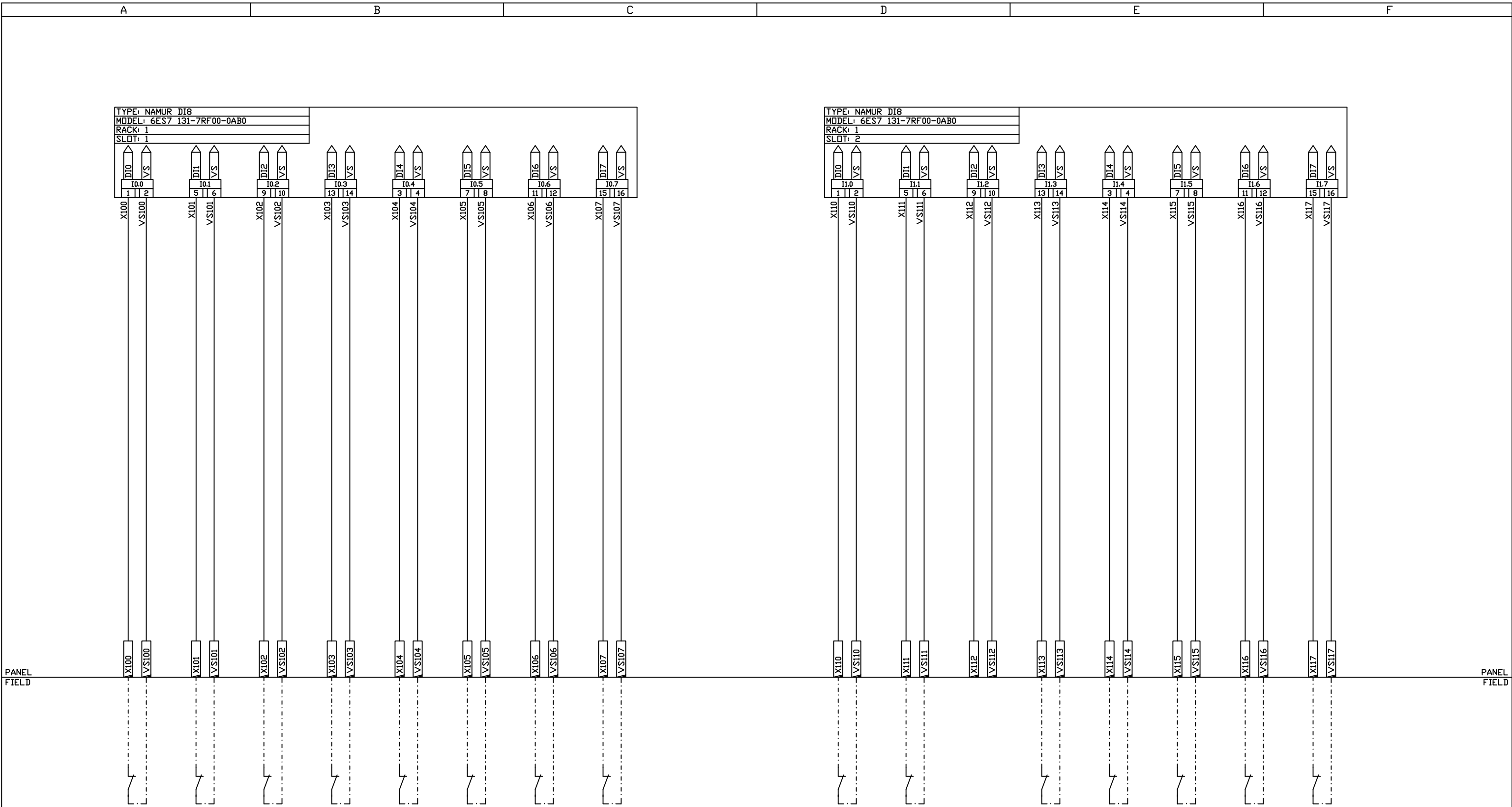


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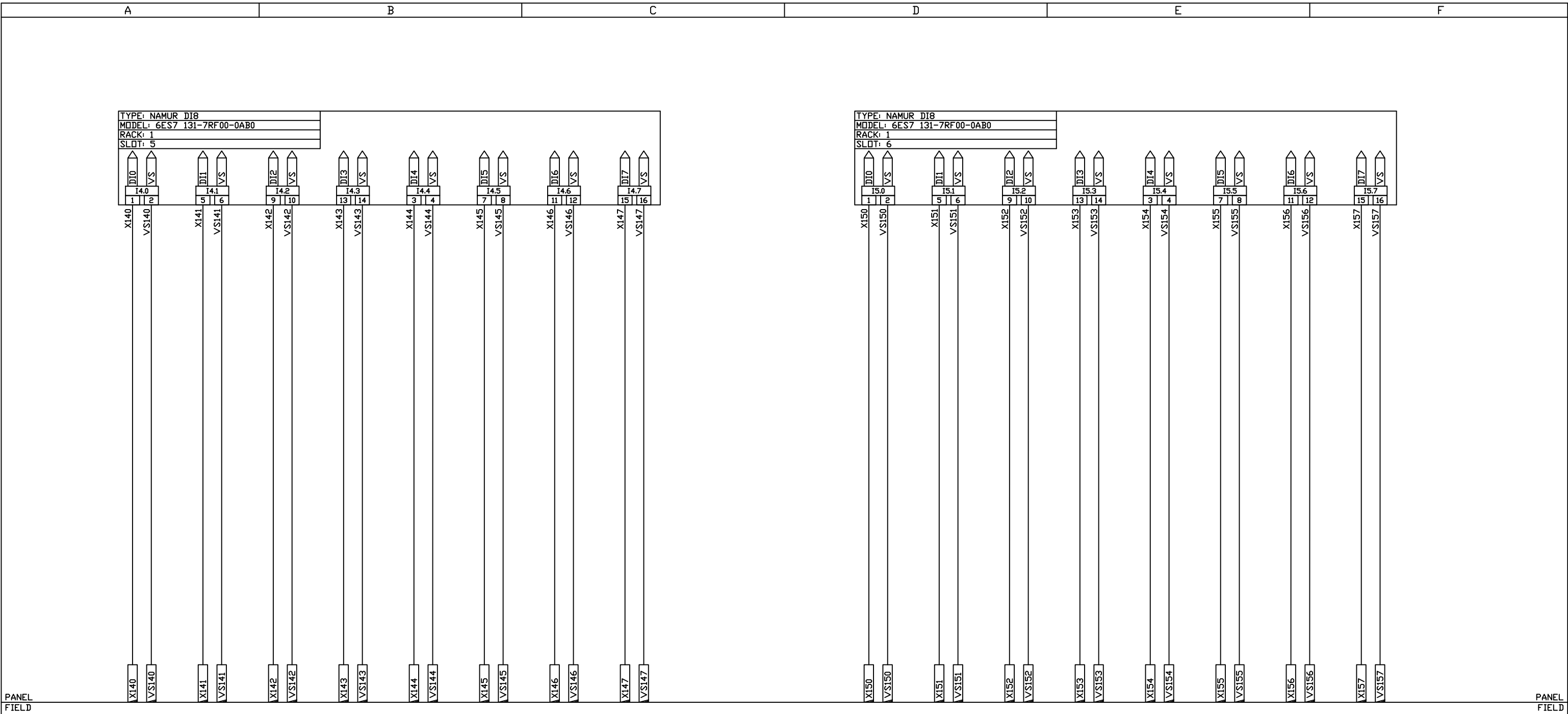


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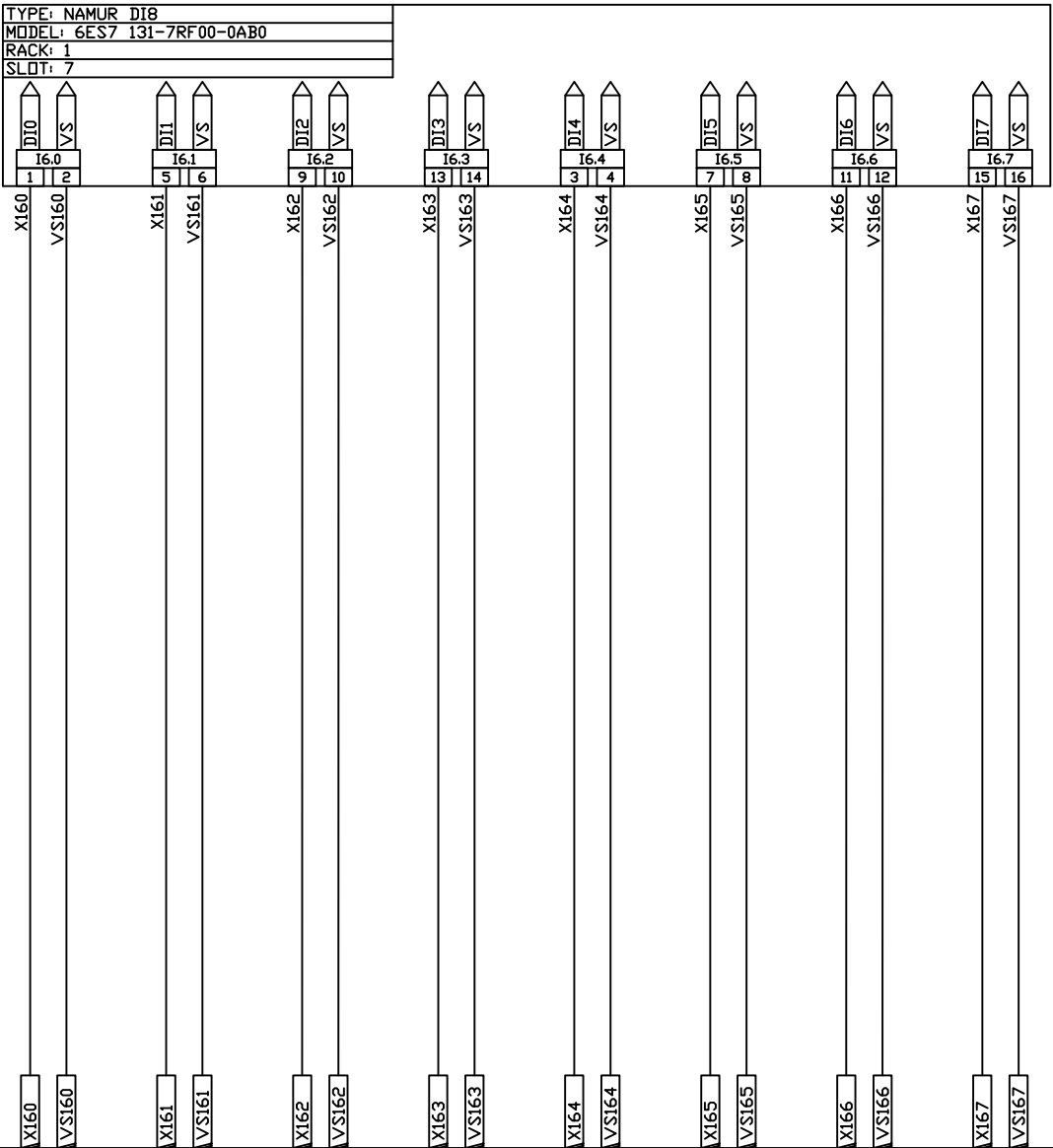


BASEMENT BELT CONVEYOR 1 BB1								BASEMENT BELT CONVEYOR 1 BB1			SPARE	BASEMENT BELT CONVEYOR 2 BB2						
BELT ALIGNMENT SENSOR TAIL LEFT	BELT RIP SENSOR TAIL LEFT	BELT ALIGNMENT SENSOR TAIL RIGHT	BELT RIP SENSOR TAIL RIGHT	BELT ALIGNMENT SENSOR HEAD LEFT	BELT RIP SENSOR HEAD LEFT	BELT ALIGNMENT SENSOR HEAD RIGHT	BELT RIP SENSOR HEAD RIGHT	GTU UPPER LIMIT	GTU LOWER LIMIT	BELT ALIGNMENT SENSOR TAIL LEFT		BELT RIP SENSOR TAIL LEFT	BELT ALIGNMENT SENSOR TAIL RIGHT	BELT RIP SENSOR TAIL RIGHT	BELT ALIGNMENT SENSOR HEAD LEFT			
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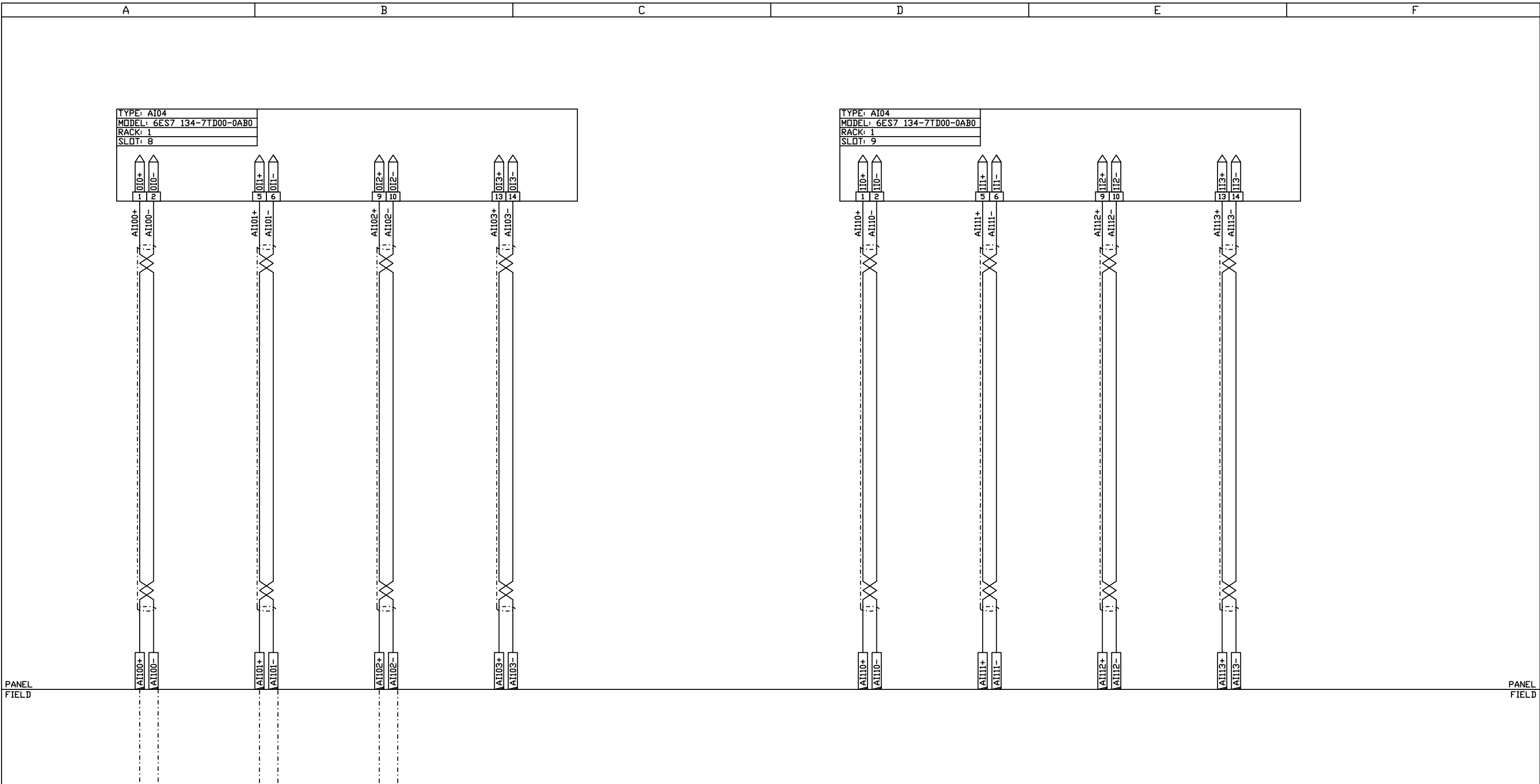
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	BB1 TEMPERATURE MONITORING		BB2 TEMPERATURE MONITORING		BB3 TEMPERATURE MONITORING		SPARE		SPARE		SPARE		SPARE		SPARE	
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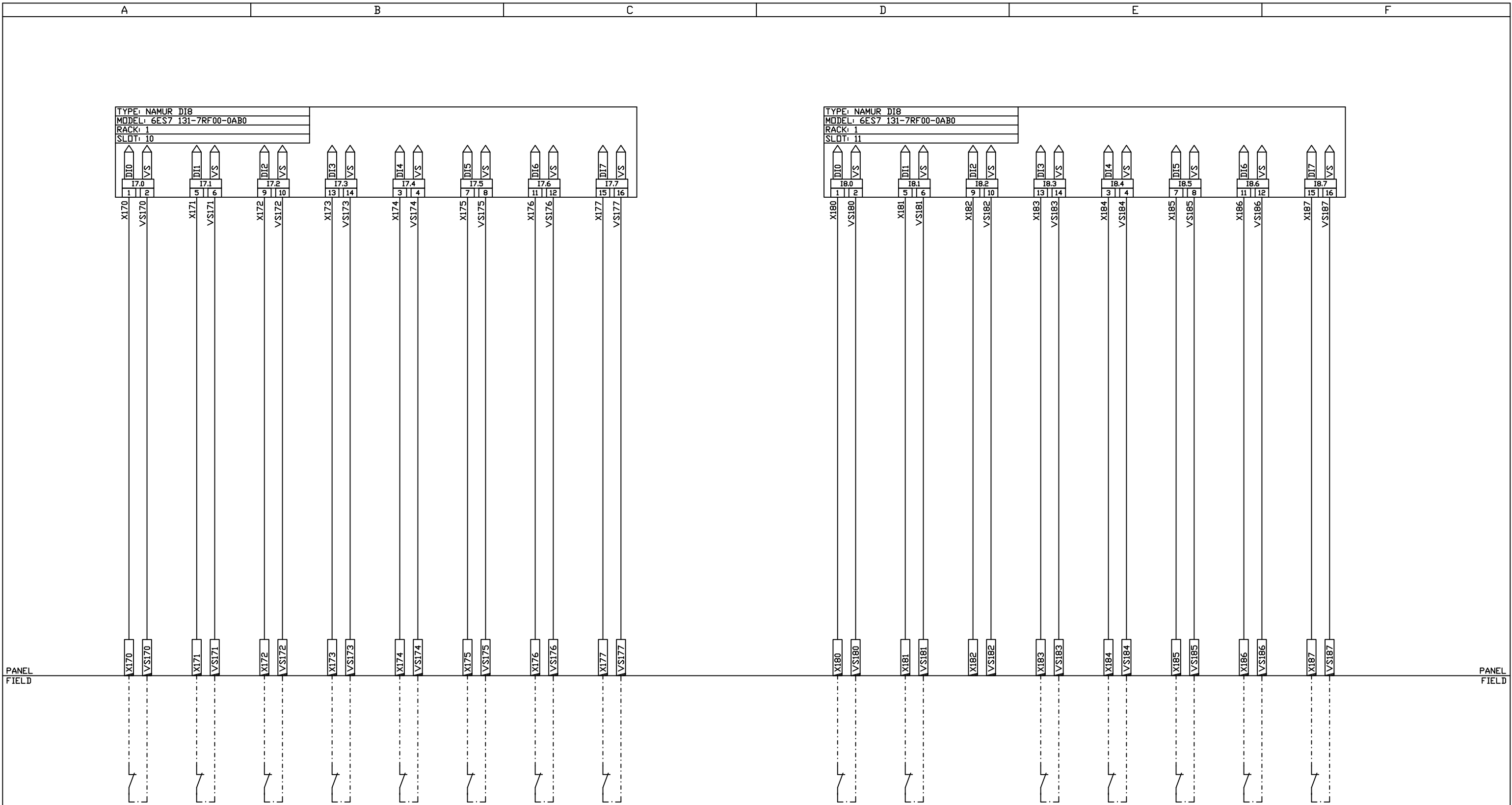
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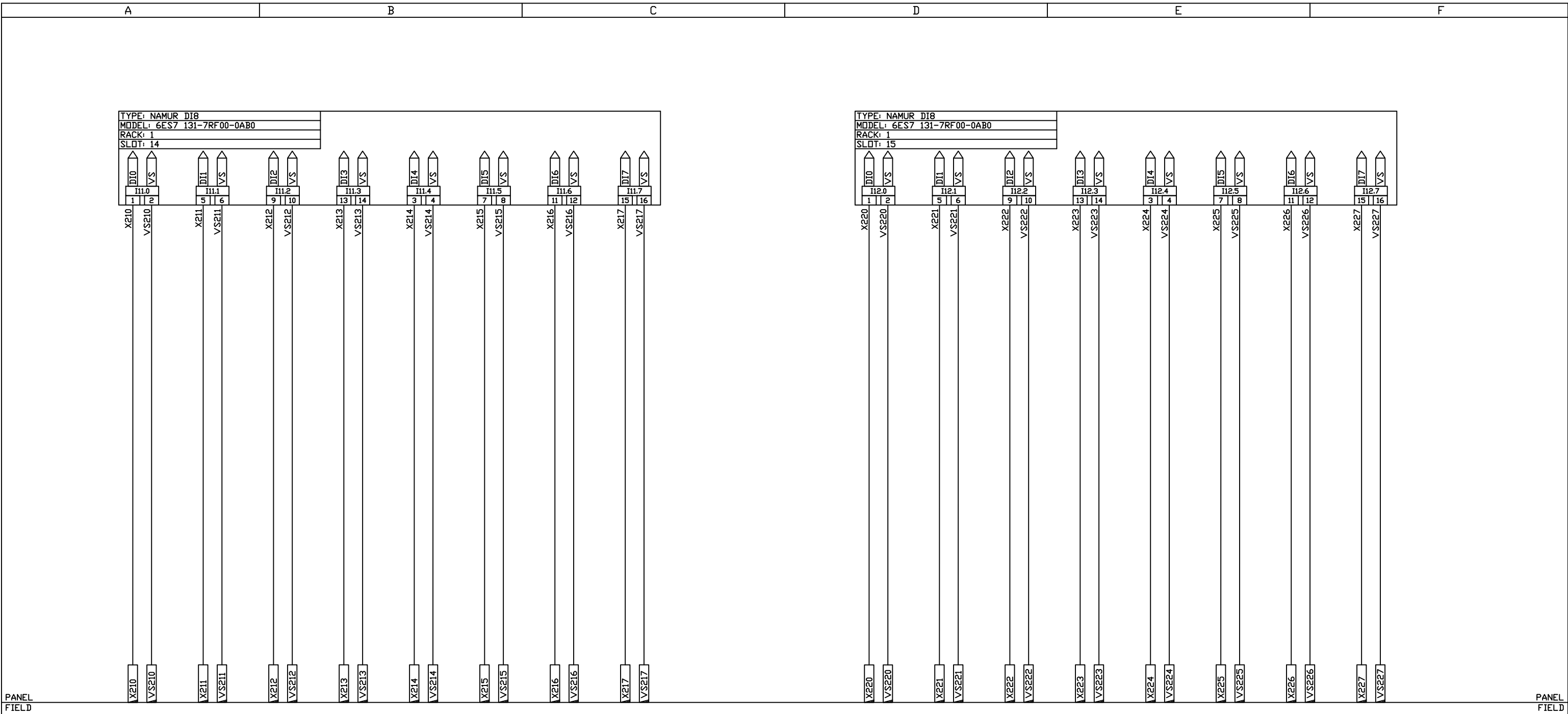


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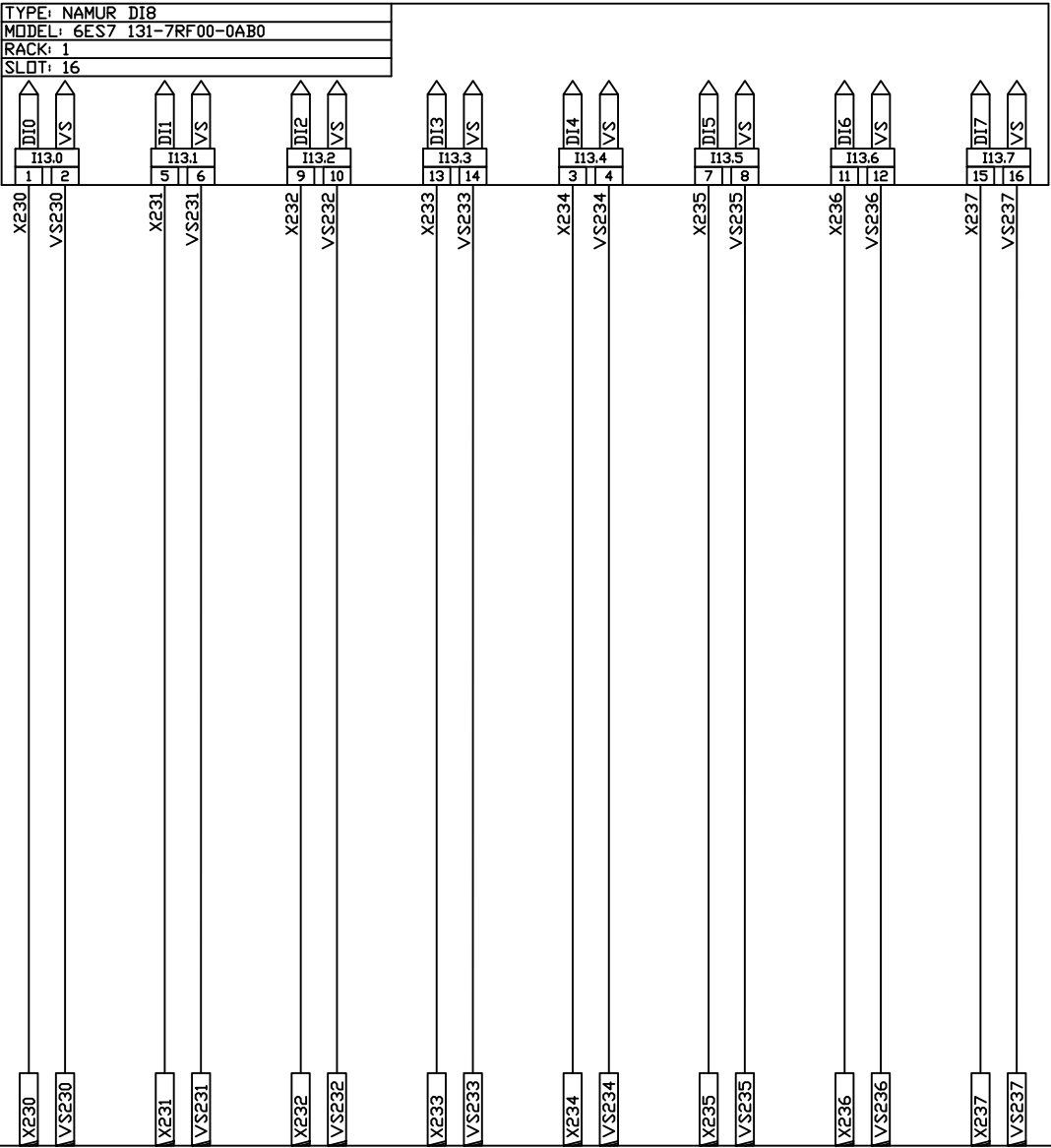




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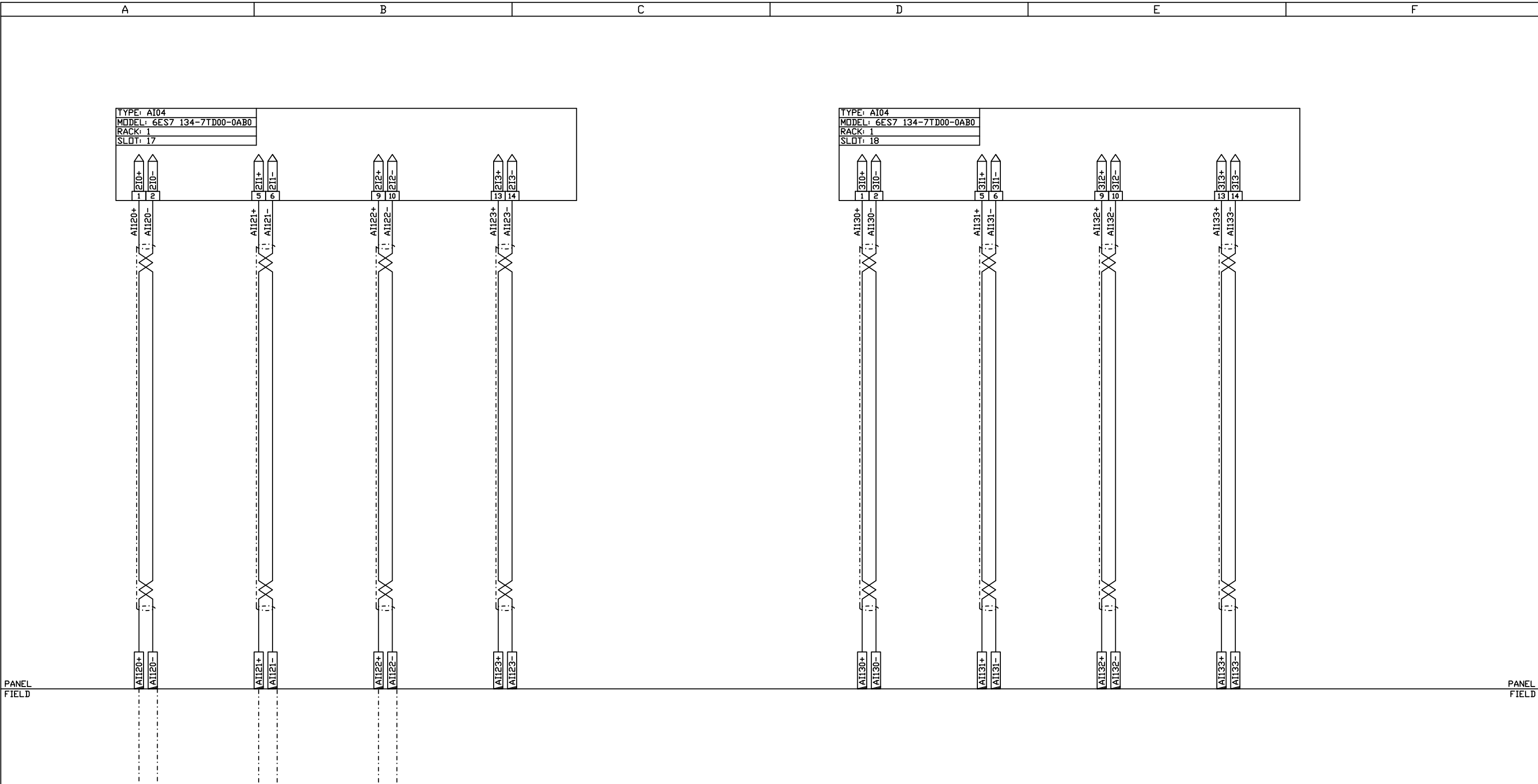
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IS PLC PANEL		00	14 OF 15
WIRING SCHEMATICS			



TYPE: AI04
MODEL: 6ES7 134-7TD00-0AB0
RACK: 1
SLOT: 18

310+

310-

311+

311-

312+

312-

313+

313-

AI130+

AI130-

AI131+

AI131-

AI132+

AI132-

AI133+

AI133-

AI130+

AI130-

AI131+

AI131-

AI132+

AI132-

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